

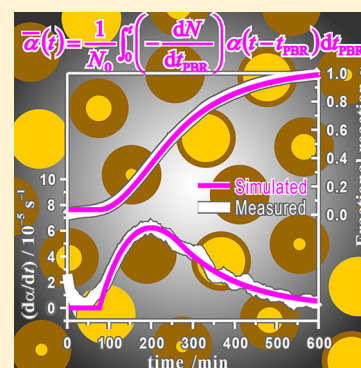
Kinetic Modeling for Thermal Dehydration of Ferrous Oxalate Dihydrate Polymorphs: A Combined Model for Induction Period–Surface Reaction–Phase Boundary Reaction

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Supporting Information

ABSTRACT: In this study, ferrous oxalate dihydrate polymorph particles, α - and β -phases, with square bipyramidal and quadratic prismatic shapes, respectively, were synthesized. Thermal dehydration of the samples was subjected to kinetic study as a typical reaction that indicates a significant induction period and a sigmoidal mass-loss behavior. On the basis of the formal kinetic analysis of the mass-loss traces recorded under isothermal, nonisothermal, and constant transformation rate conditions and the morphological observations of the surface textures of the partially reacted sample particles, a combined kinetic model for the induction period–surface reaction–phase boundary reaction was developed. The sigmoidal mass-loss behavior after the significant induction period under isothermal conditions was satisfactorily simulated by the combined kinetic model. The kinetic parameters for the component processes of induction period, surface reaction, and phase boundary reaction were separately determined from the kinetic simulation. The differences in the kinetic behaviors of the induction period and the phase boundary reaction between α - and β -phase samples were well described by the kinetic parameters. The applicability of the combined kinetic model to practical systems was demonstrated through characterizing the physicochemical kinetics of the thermal dehydration of ferrous oxalate dihydrate polymorphs.



1. INTRODUCTION

Thermal decomposition of solids is one of the most common steps during processing of a wide range of materials. The reaction mechanism and kinetics have long been studied in order to improve material syntheses and understand solid-state reactions.^{1–10} When considering the reaction of a single reactant body, such as a crystalline particle, a single crystal, or a pellet, the surfaces are the most reactive sites in many cases. Surface nucleation and subsequent propagation of solid product on the surface are the predominant processes at the beginning of a reaction. The surface reaction (SR) results in the formation of a surface product layer. The established reaction proceeds by advancement of the reaction interface, produced between the surface product layer and the internal reactant solid, toward the center of the original reactant body. If the rate of surface nucleation is instantaneous and that of the SR is greater than the advancement of the reaction interface inward, the overall kinetics are controlled by the advancement of the reaction interface and the geometry of the core shrinkage, as has been observed for thermal decomposition of selected solids under certain reaction conditions.^{11–15} However, many thermal decomposition processes are more complex.^{10,16} For instance, there might be a significant contribution of SR to the overall rate behavior,^{17,18} blocking of the diffusional removal of product gases by the surface product layer,^{19,20} changes in the reaction conditions at the reaction interface because of evolved gas and temperature fluctuations due to enthalpy

changes from the reaction,²¹ and changes in reactivity by structural phase transition²² or gelation²³ of the internal reactant solid. In addition, reaction behaviors can vary with sample properties and reaction conditions.

Thermoanalytical techniques are powerful tools for studying the rate behavior of thermal decomposition of solids.^{4,7} However, when an assemblage of sample particles is used, many other factors, besides the physicochemical kinetic behavior of a single reactant body, can influence the overall kinetics. Those factors include size distribution of the sample particles,²⁴ the temperature gradient caused by a self-cooling or self-heating effect due to enthalpy change from the reaction,²⁵ and gradient of the partial pressure of evolved gases caused by impedance of the sample particles layer on the gross diffusion of those gases.²⁶ All of these factors prevent direct correlation between gross kinetic rate data and the kinetic behavior at each reactant particle in the assemblage because of possible distribution of the degree of conversion among the particles observed during the reaction. Note that the distribution of the degree of conversion is also influenced by the characteristics of the sample particles with a distribution of the reactivity and the kinetic characteristics of the process of the SR. Accordingly, information about the physicochemical kinetics of a single

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reactant particle and assumptions about the possible gradient or distribution of the degree of conversion among the particles are necessary for a rigorous interpretation of the experimental kinetic data.

This study focuses on the kinetics of a specific type of thermal decomposition of solids, which, under isothermal conditions, evince a sigmoidal mass-loss behavior after a long induction period (IP). The existence of IPs in some thermal decomposition processes has been recognized,^{1,2} and examples can be found for many thermal decomposition and dehydration processes.^{27–29} In addition, the subsequent mass-loss process under isothermal conditions is sometimes sigmoidal.^{18,27,29} On the basis of a single reactant body, the sigmoidal mass-loss process can be ascribed to random nucleation on the surface and the subsequent isotropic growth of the nuclei toward the center of the reactant body, as formalized by Mampel's model.^{10,30–32} However, when an assemblage of sample particles is subjected to a kinetic study, a possible distribution of the IP among the particles should be considered, which imposes a distribution on the degree of conversion among the particles. Favergeon et al. reported experimental evidence for the distribution of the IP among single crystals for the thermal dehydration of lithium sulfate monohydrate.²⁸ Yoshioka et al. proposed a kinetic model for the two-dimensional phase-boundary-controlled reaction, with a distribution of the reaction initiation time among the reactant particles in the sample assemblage, for describing the overall kinetics of the thermal decomposition of magnesium hydroxide.³³ This model was later adopted by Masuda et al. to study the thermal dehydration of magnesium oxalate dihydrate.³⁴ A similar model was also derived by Pijolat et al. to describe the kinetics of the thermal decomposition and solid–gas reactions of a single reactant particle as the nucleation and anisotropic growth model.^{35–37}

In this study, the thermal dehydration of ferrous oxalate dihydrate polymorphs was selected for the kinetic study because the experimental kinetic rate data under isothermal conditions indicate a sigmoidal mass-loss behavior after a long IP.³⁸ The thermal dehydration process of ferrous oxalate dihydrate has been studied by many researchers in order to understand the synthetic process and control the morphology of iron(III) oxide polymorphs via this thermal decomposition process.^{38–53} We investigated the reaction kinetics through the formal kinetic analysis of the experimental mass-loss data recorded under different heating program modes and the morphological observations of the reacting particles. On the basis of the findings, we derived an overall kinetic model that integrates the kinetics of (1) the IP, (2) random nucleation on the surface, and (3) advancement of the reaction interface. The practical applicability of the kinetic model and its merits were demonstrated by comparing the results of kinetic simulations for the thermal dehydration of the two polymorph phases of ferrous oxalate dihydrate.

2. EXPERIMENTAL SECTION

2.1. Sample Preparation. Aqueous solutions of oxalic acid and ferrous ammonium sulfate of 0.1 mol dm^{−3} each were prepared by dissolving, respectively, oxalic acid dihydrate (Special grade, Sigma–Aldrich Japan) and ferrous ammonium sulfate hexahydrate (Special grade, Sigma–Aldrich Japan) into deionized–distilled water. The ferrous ammonium sulfate solution (250 cm³) was poured into the oxalic acid solution (300 cm³) in an Erlenmeyer flask, and the mixture was heated to 363 K in a water bath. The mixed solution with precipitates

was mechanically stirred for different times of 1–12 h by keeping the temperature at 363 K. The precipitate was then separated by vacuum filtration and repeatedly rinsed with deionized–distilled water. The separated solids were dried in a vacuum desiccator for 24 h and stored in a refrigerator at 278 K. The samples prepared by stirring for different times were sorted out by designating according to the stirring time; for example, the sample stirred for 1 h was designated as the 1h sample.

2.2. Sample Characterization. The samples were characterized with powder X-ray diffraction (XRD), Fourier transfer infrared spectroscopy (FTIR), and thermogravimetry–differential thermal analysis (TG–DTA). For the samples press-fitted to a glass plate, the XRD pattern was recorded using a diffractometer (RINT2200 V, Rigaku Co.) with monochrome Cu K α radiation (40 kV, 20 mA). FTIR spectra of the samples diluted with KBr were measured by the diffuse reflectance method using a spectrophotometer (FT-IR8400S, Shimadzu Co.). TG–DTA curves were recorded using a simultaneous TG–DTA thermal analyzer (DTG50, Shimadzu) on 5.0 mg of sample weighed into a platinum cell (6 mm ϕ , 2.5 mm height) at a heating rate β of 10 K min^{−1} in flowing air at a rate of 80 cm³ min^{−1}. The morphology of the samples was observed using a scanning electron microscope (SEM; JSM-6510, Jeol) after coating the sample with platinum by sputtering. The specific surface area of the selected samples was measured by a single-point BET method using a surface area analyzer (FlowSorbII-2300, Micromeritics Co.) after a pretreatment of these samples at 373 K in flowing N₂ for 1 h. The sample particles were dispersed in deionized–distilled water with a small amount of surfactant (sodium hexametaphosphate, Sigma–Aldrich Japan), and the particle size distribution was evaluated using a laser diffraction particle size analyzer (SALD-300 V, Shimadzu Co.).

2.3. Measurements of Thermal Dehydration Behavior. The 1h and 12h samples were press-fitted onto platinum plates without grinding. Changes of the XRD patterns of these samples during the thermal dehydration under isothermal conditions were tracked at 423 and 433 K, respectively, in flowing N₂ (100 cm³ min^{−1}). The XRD patterns were recorded repeatedly using RINT2200 V equipped with a programmable heating chamber (PTC-20A, Rigaku Co.). The diffraction measurements were started every 15 min by keeping a constant temperature.

To characterize the morphological changes during the thermal dehydration process, the samples were heated in DTG-50 with flowing N₂ (80 cm³ min^{−1}) at constant temperatures (403 K for the 1h sample, 412 and 415 K for the 12h sample) for different durations, depending on the degree of dehydration desired. Then, they were cooled to room temperature and observed with a SEM.

To trace the thermal dehydration behavior of the 1h and 12h samples ($m_0 = 5.0$ mg, weighed in a platinum cell (6.0 mm ϕ and 2.5 mm height)), mass-loss traces under different heating program modes (i.e., isothermal, linear nonisothermal, and controlled transformation rate (CR) modes) in flowing N₂ (80 cm³ min^{−1}) were collected using a hanging-type TG instrument (TGA-50, Shimadzu Co.). The isothermal mass-loss traces were recorded at different constant temperatures in a range of 394–412 K for the 1h sample and 403–418 K for the 12h sample. These constant temperatures were reached by heating the samples at a β of 10 K min^{−1}. The mass-loss traces under linearly increasing temperature were measured at different β , 0.5 $\leq \beta \leq 5.0$ K min^{−1}. For the CR measurements, a

homemade controller of the sample-controlled thermal analysis (SCTA)^{54,55} was attached to the TGA-50 instrument, which would adjust the temperature to achieve a programmed constant mass-loss rate C . Using this experimental setup, temperature profiles to achieve different C , ranging from 5.0 to 15 $\mu\text{g min}^{-1}$, were recorded.

3. RESULTS AND DISCUSSION

3.1. Formation of Polymorphs. Figure 1 shows the XRD patterns of the 1h–12h samples. There was a change in the

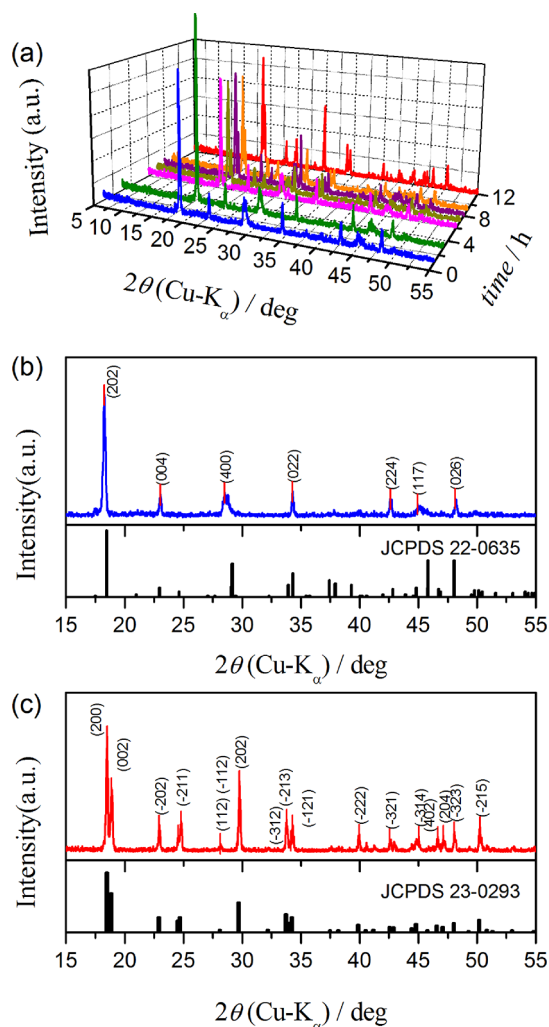


Figure 1. XRD patterns of samples: (a) 1h–12h, (b) 1h, and (c) 12h.

XRD pattern between the 6h and 9h samples, and the degree of the change increased with aging duration at 363 K (Figure 1a). The XRD pattern of the 1h sample (Figure 1b) corresponds to that reported for $\beta\text{-FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (orthorhombic, $Cccm$, $a = 12.26$, $b = 5.57$, $c = 15.48$).⁵⁶ The XRD pattern changes to that corresponding to $\alpha\text{-FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (monoclinic, $C2/c$, $a = 12.05$, $b = 5.57$, $c = 9.76$, $\beta = 124^\circ 18'$)⁵⁶ in the 12h sample (Figure 1c). The phase transformation behavior with aging of the initial precipitates in the hot mother solution with mechanical stirring is in accordance with that in a previous report.⁵² FTIR absorption spectra of all samples were nearly identical irrespective of the aging treatment and coincided with those reported for metal oxalate hydrates, with absorption

peaks due to crystalline water, the oxalate ion, and the metal–O bond in metal oxalates.^{49,57–59}

Figure 2 compares the TG–DTG–DTA curves for the 1h and 12h samples in flowing air. The 1h and 12h samples had total mass-

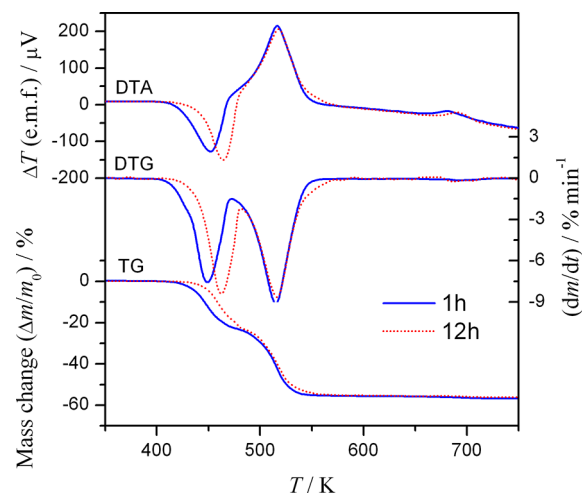
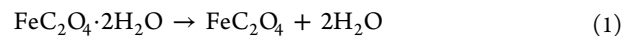


Figure 2. Comparisons of TG–DTG–DTA curves for the 1h and 12h samples ($m_0 = 5.0$ mg) at $\beta = 10$ K min^{-1} in flowing air (80 $\text{cm}^3 \text{min}^{-1}$).

loss values of 56.8 and 57.1%, respectively. It has been reported that the reaction proceeds via two mass-loss steps of thermal dehydration and oxidative decomposition of anhydride in air.^{39–44,47}



The observed total mass-loss value is in good agreement with the calculated value, based on eqs 1–4. The thermal dehydration and oxidative decomposition processes show the endothermic and exothermic effects, respectively. Specifically, the exothermic behavior during the second mass-loss step resulted from the oxidation of the as-produced CO in eq 3 and the oxidation of FeO in eq 4. The subsequent exothermic effect initiates at around 600 K and is accompanied by a detectable mass loss, corresponding to the crystallization of $\alpha\text{-Fe}_2\text{O}_3$ from an intermediate amorphous product.^{38,51,52} Comparing the thermal behavior of the 1h and 12h samples, distinct differences are seen in the reaction temperatures and shapes of TG and DTG curves for the thermal dehydration processes, but the rate behavior for the oxidative decomposition (second mass-loss step) and the subsequent crystallization process is nearly identical between the samples.

Figure 3 shows typical SEM images and particles size distributions for the 1h (β -phase) and 12h (α -phase) samples. The specific surface areas of these samples were 1.54 ± 0.08 and $0.85 \pm 0.03 \text{ m}^2 \text{g}^{-1}$, respectively. The particles in the 1h and 12h samples are characterized as quadratic prisms and square bipyramidals, respectively. The morphological characteristics of the samples are in accordance with those reported previously for the polymorphs.^{38,52} A relatively large distribu-

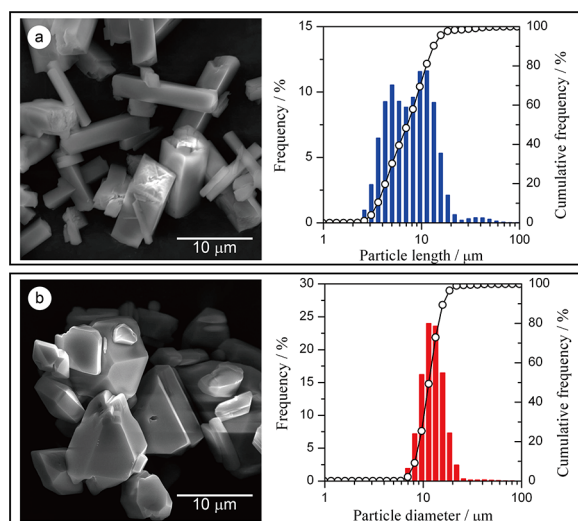


Figure 3. Typical SEM images and particle size distribution of the samples: (a) 1h and (b) 12h.

tion of the particle length of the 1h sample (β -phase) is observed with an average value of $7.09 \pm 0.23 \mu\text{m}$. The average particle diameter of the 12h sample (α -phase), $10.73 \pm 0.11 \mu\text{m}$, is larger than that of the 1h sample. The differences in morphology and particle size between these samples, in addition to the crystallographic difference, could have affected the rate behavior of the thermal dehydration. The 1h and 12h samples were used to study the thermal dehydration kinetics of β - and α -phases, respectively.

3.2. Comparison of Thermal Dehydration Processes.

Figures 4 and 5 illustrate the mass-change traces for the thermal dehydration of the α - and β -phases, respectively. Irrespective of the reaction temperature conditions under isothermal, linear nonisothermal, and CR conditions, both samples show smooth mass-loss traces. Under isothermal conditions, a distinguishable IP for the thermal dehydration is observed for both α - and β -phases, and the mass-loss traces are sigmoidal. A significant temperature drop at an early stage in each reaction is observed for the temperature profiles of the mass-loss measurements under CR conditions. This behavior is characteristic of a reaction controlled by the surface nucleation process.^{60,61} After reaching the minimum temperature, the reaction temperature gradually increases with an accelerating increase in the final part of the reaction. The reaction temperature profile in the latter half of the reaction is typical for a reaction controlled by interface shrinkage.^{60,61}

Figure 6 shows changes in the XRD pattern with reaction time during heating of the β -phase sample at a constant temperature. The diffraction peaks attributed to the β -phase gradually attenuate as the reaction advances (Figure 6a). The product of thermal dehydration is poorly crystalline anhydrous ferrous oxalate (Figure 6b). For the α -phase sample, the XRD patterns indicated a similar attenuation of the hydrated phase and growth of poorly crystalline anhydrous ferrous oxalate, without any intermediate crystalline phase.

3.3. IP. The IP observed in the mass-loss traces under isothermal conditions could be correlated with the kinetic process for the formation of possible nucleus-forming sites on the surfaces of the sample particles.^{6,27,29} Operationally, the IPs at different temperatures were determined as the time interval between the initial achievement point to the constant temperature and the onset of the derivative mass-loss trace

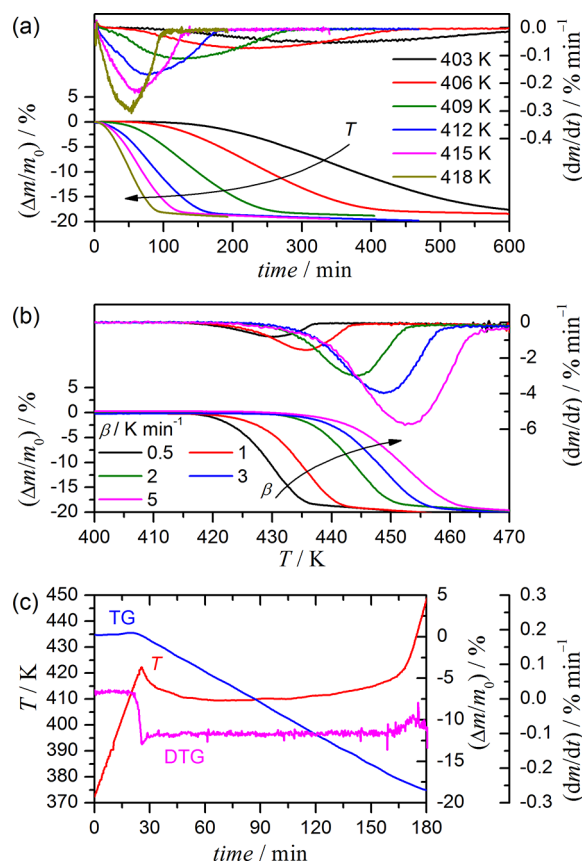


Figure 4. Mass-change traces for the thermal dehydration of α - $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ ($m_0 = 5.0 \text{ mg}$) in flowing N_2 ($80 \text{ cm}^3 \text{ min}^{-1}$): (a) isothermal, (b) nonisothermal, and (c) constant transformation rate conditions.

for the thermal dehydration. Figure 7a shows the temperature-dependent change in the IP. When the Arrhenius-type temperature dependence of the IP is assumed, the apparent kinetic behavior in the IP can be expressed by the kinetic equations^{27,29}

$$\frac{d\alpha_{\text{IP}}}{dt} = k_{\text{IP}} f(\alpha_{\text{IP}}) \quad \text{and} \quad k_{\text{IP}} = A_{\text{IP}} \exp\left(-\frac{E_{\text{IP}}}{RT}\right) \quad (5)$$

where α_{IP} , k_{IP} , and $f(\alpha_{\text{IP}})$ are the saturation degrees of the nucleus-forming site, the apparent rate constant for the formation of the nucleus-forming site, and the conversion function, which describes the mechanistic characteristics of the process, respectively. A_{IP} and E_{IP} are the Arrhenius pre-exponential factor and apparent activation energy, respectively, for the IP. R is the gas constant. The reciprocal value of the IP, t_{IP} , is equivalent to the average formation rate of nucleus-forming sites, $(d\alpha_{\text{IP}}/dt)_{\text{avg}}$. At a point of $(d\alpha_{\text{IP}}/dt)_{\text{avg}}$ during the IP, the value of α_{IP} should be a constant irrespective of the temperature. These relationships lead to the following equation for analyzing the apparent kinetics of the IP through an isoconversional plot.

$$\ln\left(\frac{1}{t_{\text{IP}}}\right) = \ln[A_{\text{IP}} f(\alpha_{\text{IP}})] - \frac{E_{\text{IP}}}{RT} \quad (6)$$

Through plotting $\ln(1/t_{\text{IP}})$ against T^{-1} , the validity of the assumption of the Arrhenius-type temperature dependence is

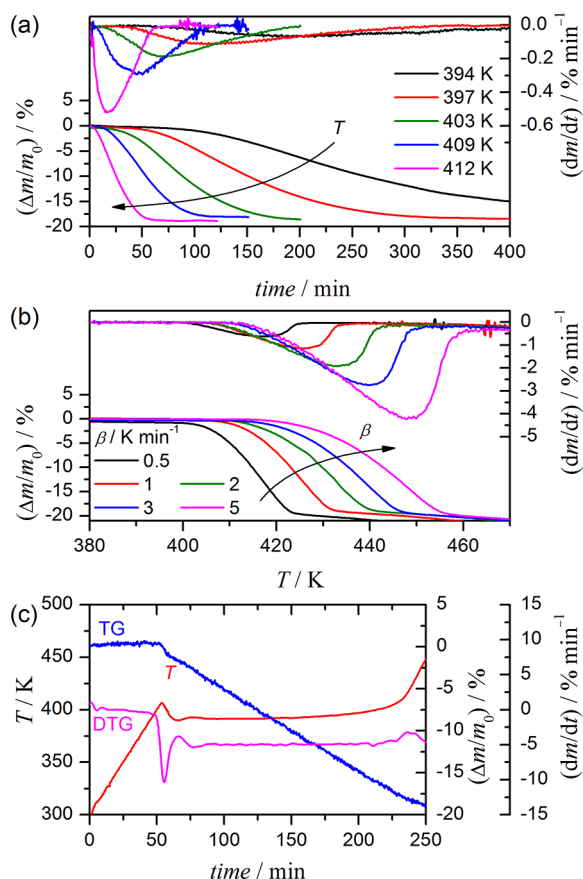


Figure 5. Mass-change traces for the thermal dehydration of β - $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ ($m_0 = 5.0$ mg) in flowing N_2 ($80 \text{ cm}^3 \text{ min}^{-1}$): (a) isothermal, (b) nonisothermal, and (c) constant transformation rate conditions.

evaluated from the linearity of the plot, and the E_{IP} value can be obtained from the slope.^{27,29}

In Figure 7b, plots of $\ln(1/t_{\text{IP}})$ versus T^{-1} for the IPs for thermal dehydration of the α - and β -phases are compared. Both plots indicate a linear relationship for $\ln(1/t_{\text{IP}})$ versus T^{-1} . The apparent values of E_{IP} for the α - and β -phases were calculated as 352.4 ± 29.2 and $235.8 \pm 8.0 \text{ kJ mol}^{-1}$, respectively. A higher-energy barrier for nucleus site formation is expected for the α -phase because of the longer IP at a constant temperature and the larger apparent E_{IP} value as compared to the β -phase.

3.4. Formal Kinetics of Thermal Dehydration. For describing the formal kinetics of the thermal dehydration process, the following fundamental kinetic equation was assumed to be applied under all temperature conditions⁶²

$$\frac{d\alpha}{dt} = A \exp\left(-\frac{E_a}{RT}\right) f(\alpha) \quad (7)$$

where A and E_a are the apparent values for the Arrhenius pre-exponential factor and the apparent activation energy for the thermal dehydration process, respectively. $f(\alpha)$ is the kinetic model function derived by assuming physicochemical reaction mechanisms. The logarithmic form of eq 7 is expressed as

$$\ln\left(\frac{d\alpha}{dt}\right) = \ln[Af(\alpha)] - \frac{E_a}{RT} \quad (8)$$

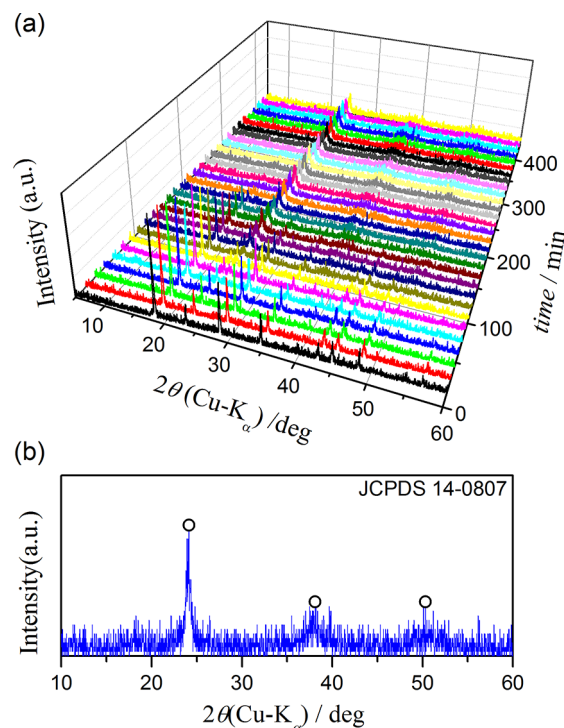


Figure 6. Changes in the XRD pattern for the isothermal dehydration of β - $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ at 423 K in flowing N_2 ($100 \text{ cm}^3 \text{ min}^{-1}$) (a) and the XRD pattern of the final dehydration product (b).

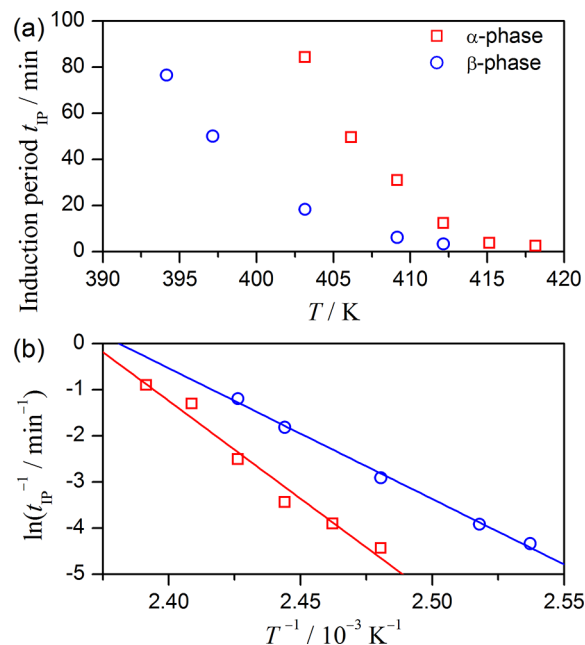


Figure 7. Kinetic analysis for the IP: (a) change in the IP with reaction temperature and (b) Arrhenius plots for the IP.

At a fixed α , the plot of $\ln(d\alpha/dt)$ versus T^{-1} , known as the Friedman plot,⁶³ should be linear because $\ln[Af(\alpha)]$ should be constant. This isoconversional relationship can be evaluated by selecting data points at a fixed α for the kinetic rate data under different temperature conditions, that is, for isothermal, linear nonisothermal, and CR conditions. Figure 8 shows the results of the isoconversional analysis. The Friedman plots are linear irrespective of the phases or the α value selected (Figure 8a and

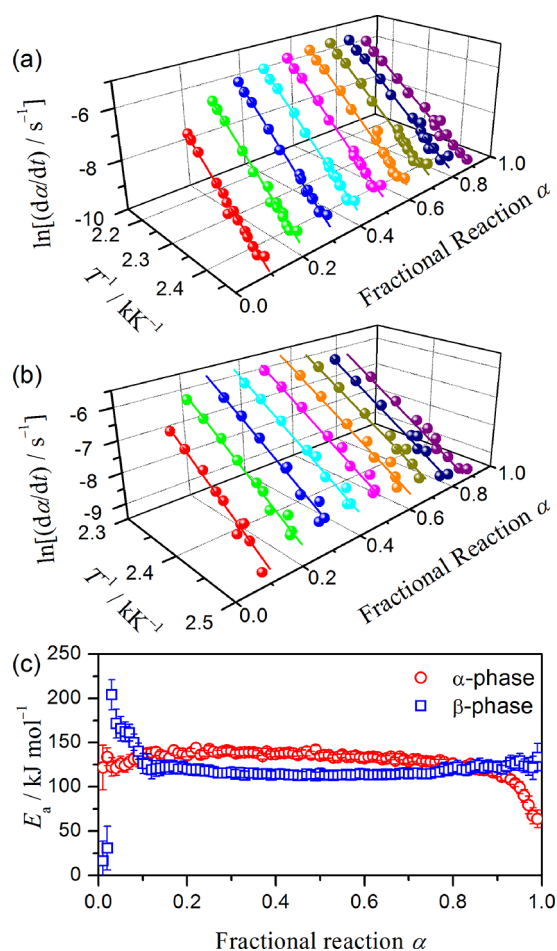


Figure 8. Results of isoconversional kinetic analysis for the thermal dehydration of α - and β - $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$: (a) Friedman plots for the α -phase, (b) Friedman plots for the β -phase, and (c) apparent E_a values at different α values.

b). The apparent E_a values at different α values calculated from the slopes of the Friedman plots indicate constant values during the main part of the reaction ($0.1 \leq \alpha \leq 0.9$) for both α - and β -phases (Figure 8c). The average apparent E_a values ($0.1 \leq \alpha \leq 0.9$) for the reaction of α - and β -phases were 133.4 ± 6.5 and 116.8 ± 3.7 kJ mol^{-1} , respectively. The average apparent E_a value for the β -phase is in good agreement with the value reported for the isothermal dehydration of the β -phase using a different preparation, $E_a = 110.4 \pm 5.4$ kJ mol^{-1} ($0.1 \leq \alpha \leq 0.9$).³⁸

Supported by the experimental evidence of acceptable constant apparent E_a values during the main part of the reaction, the experimental master plot of $d\alpha/d\theta$ versus α could be drawn by extrapolating the kinetic rate data to infinite temperature according to the following equation^{62,64–66}

$$\frac{d\alpha}{d\theta} = \left(\frac{d\alpha}{dt} \right) \exp\left(\frac{E_a}{RT} \right) \quad \theta = \int_0^t \exp\left(-\frac{E_a}{RT} \right) dt \quad (9)$$

where θ is the generalized time proposed by Ozawa,^{62,67,68} which denotes the hypothetical reaction time at infinite temperature. Figure 9 shows the experimental master plots for the thermal dehydration of the α - and β -phases.

The shape of the experimental master plot can be correlated with the kinetic model function^{62,64–66}

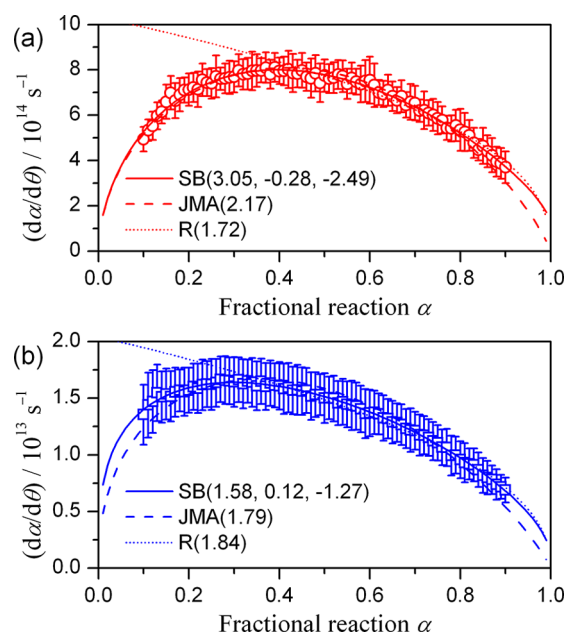


Figure 9. Experimental master plots of $d\alpha/d\theta$ versus α for the thermal dehydration processes and the fitting curves using $\text{SB}(m,n,p)$, $\text{JMA}(m)$, and $\text{R}(n)$ functions: (a) α -phase and (b) β -phase.

$$\frac{d\alpha}{d\theta} = Af(\alpha) \quad (10)$$

Both of the experimental master plots indicate the maxima for $d\alpha/d\theta$ during the reaction. This is a typical rate behavior for nucleation and growth or an autocatalytic reaction. We then attempted to fit the experimental master plots using both an empirical kinetic model function known as the Šesták–Berggren model,^{69,70} $\text{SB}(m,n,p)$, and the nucleation–growth-type model,^{71–74} $\text{JMA}(m)$.

$\text{SB}(m,n,p)$:

$$f(\alpha) = \alpha^m(1-\alpha)^n[-\ln(1-\alpha)]^p \quad (11)$$

$\text{JMA}(m)$:

$$f(\alpha) = m(1-\alpha)[- \ln(1-\alpha)]^{1-1/m} \quad (12)$$

By allowing nonintegral values for the kinetic exponents in eqs 11 and 12, these equations can more accurately fit the experimental master plots and accommodate possible deviations of the actual reaction process from the ideal model.^{75,76} For both α - and β -phase samples, $\text{JMA}(m)$ with $m \approx 2$ fit the experimental master plots well. However, $\text{JMA}(m)$ curves are slightly different from the empirical $\text{SB}(m,n,p)$ curves, which exhibit a nearly perfect fit. For the α -phase, the deviations are observed in the final part of the reaction, and for the β -phase, deviations are seen in both initial and final parts of the reaction. To better fit the latter half of the reaction with the kinetic model function, a phase-boundary-controlled model $\text{R}(n)$ ⁷⁷ was applied for the range $0.30 \leq \alpha \leq 0.90$.

$\text{R}(n)$:

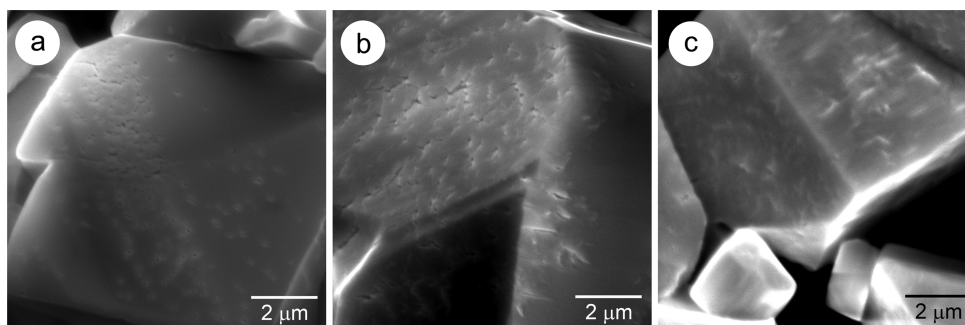
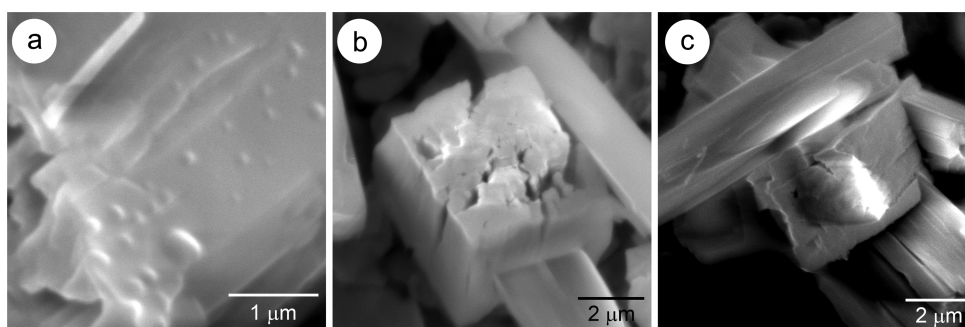
$$f(\alpha) = n(1-\alpha)^{1-1/n} \quad (13)$$

For both phases, the decelerating part of the reactions can be perfectly fitted using $\text{R}(n)$ functions with $n \approx 2$. Table 1 summarizes the kinetic results obtained by the formal kinetic analysis.

Table 1. Summary of the Kinetic Parameters Determined by the Formal Kinetic Analysis for the Thermal Dehydration of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ Polymorphs

sample	$E_a/\text{kJ mol}^{-1}$	$f(\alpha)$ (α range)	kinetic exponents	A/s^{-1}	R^{2a}
α -phase	133.4 ± 6.5 ($0.1 \leq \alpha \leq 0.9$)	SB(m,n,p) ($0.1 \leq \alpha \leq 0.9$)	$m = 3.05 \pm 0.53$	$(2.13 \pm 0.09) \times 10^{15}$	0.9846
			$n = -0.28 \pm 0.20$		
			$p = -2.49 \pm 0.51$		
β -phase	116.8 ± 3.7 ($0.1 \leq \alpha \leq 0.9$)	JMA(m) ($0.1 \leq \alpha \leq 0.9$)	$m = 2.17 \pm 0.03$	$(8.90 \pm 0.09) \times 10^{14}$	0.9587
		R(n) ($0.3 \leq \alpha \leq 0.9$)	$n = 1.72 \pm 0.03$	$(6.01 \pm 0.06) \times 10^{14}$	0.9702
		SB(m,n,p) ($0.1 \leq \alpha \leq 0.9$)	$m = 1.58 \pm 0.24$	$(3.11 \pm 0.07) \times 10^{13}$	0.9979
			$n = 0.13 \pm 0.09$		
			$p = -1.27 \pm 0.23$		
		JMA(m) ($0.1 \leq \alpha \leq 0.9$)	$m = 1.79 \pm 0.03$	$(2.08 \pm 0.03) \times 10^{13}$	0.9471
		R(n) ($0.3 \leq \alpha \leq 0.9$)	$n = 1.84 \pm 0.02$	$(1.11 \pm 0.01) \times 10^{13}$	0.9901

^aDetermination coefficient for the nonlinear least-squares fitting of $dx/d\theta$ versus α plot.

**Figure 10.** Surface textures of $\alpha\text{-FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ partially dehydrated for different times at constant temperatures in flowing N_2 : (a) at 412 K for 5 min ($\alpha = 0.05$), (b) at 415 K for 30 min ($\alpha = 0.45$), and (c) at 415 K for 400 min ($\alpha = 0.85$).**Figure 11.** Surface textures of $\beta\text{-FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ partially dehydrated for different times at constant temperatures in flowing N_2 : (a) at 403 K for 10 min ($\alpha = 0.10$), (b) at 403 K for 30 min ($\alpha = 0.55$), and (c) at 403 K for 250 min ($\alpha = 0.85$).

The JMA(m) model was originally derived for random nucleation and growth-type reactions,^{71–74} where the expansion of the reaction interface from the initial site of the reaction and its subsequent overlapping with neighboring reaction interfaces were assumed. Although acceptable fittings of the JMA(m) model to sigmoidal mass-loss curves have been reported for thermal dehydration and decomposition of many solids, the apparent fit does not indicate the direct correspondence of the physicochemical mechanism of the reaction and the assumptions in the JMA(m) model.⁷⁸ For example, sigmoidal mass-loss traces for the thermal dehydration and decomposition have been observed for the reactions where nucleation and growth processes on particle surfaces play predominant roles in the early stages of the reaction.^{17,18} Significantly long IPs observed in many thermal dehydration and decomposition reactions^{27–29} could be signs of the surface-reaction-induced processes. The long IPs observed in this study indicate a possible contribution of SR to the overall reaction

kinetics. The possibility of surface nucleation and growth processes at the beginning of the reaction is also supported by observations of temperature decreases during that reaction stage in the measured temperature profiles for mass-loss traces in the CR mode.

However, mass-loss traces recorded using TG are an average over all particles in the sample assemblage; therefore, the possible distribution of the initiation time of the SR among the sample particles must be taken into account. The variation in reaction initiation time among the particles causes a distribution in the degree of conversion for the particles for the subsequent course of the reaction. There can also be a distribution in the degree of conversion among the particles when a significant distribution in particle size exists.²⁴ It has been proved through mathematical simulations that, even if there is significant distribution in the degree of conversion among the particles, the apparent E_a values evaluated by an isoconversional method describe the linear advancement rate of

Table 2. Differential Kinetic Equations of the Combined IP–SR–PBR Model with the Dimension n for the Reaction Interface Shrinkage^a

n	Kinetic Equation: $\frac{d\alpha}{dt} =$	Eq.
1	a) $t - t_{IP} \leq 1/k_{PBR}$: $k_{PBR} [1 - \exp(-k_{SR}(t - t_{IP}))]$	(14)
	b) $t - t_{IP} \geq 1/k_{PBR}$: $k_{PBR} \exp(-k_{SR}(t - t_{IP})) \left[\exp\left(\frac{k_{SR}}{k_{PBR}}\right) - 1 \right]$	(15)
2	a) $t - t_{IP} \leq 1/k_{PBR}$: $-2k_{PBR} \left[\left(1 + \frac{k_{PBR}}{k_{SR}}\right) \exp(-k_{SR}(t - t_{IP})) + k_{PBR}(t - t_{IP}) - \left(1 + \frac{k_{PBR}}{k_{SR}}\right) \right]$	(16)
	b) $t - t_{IP} \geq 1/k_{PBR}$: $-2k_{PBR} \exp(-k_{SR}(t - t_{IP})) \left[1 + \frac{k_{PBR}}{k_{SR}} - \frac{k_{PBR}}{k_{SR}} \exp\left(\frac{k_{SR}}{k_{PBR}}\right) \right]$	(17)
3	a) $t - t_{IP} \leq 1/k_{PBR}$: $-3k_{PBR} \left[\left(1 + 2\frac{k_{PBR}}{k_{SR}} + 2\left(\frac{k_{PBR}}{k_{SR}}\right)^2\right) \exp(-k_{SR}(t - t_{IP})) - (k_{PBR}(t - t_{IP}))^2 + 2k_{PBR}\left(\frac{k_{PBR}}{k_{SR}} + 1\right)(t - t_{IP}) - \left(1 + 2\frac{k_{PBR}}{k_{SR}} + 2\left(\frac{k_{PBR}}{k_{SR}}\right)^2\right) \right]$	(18)
	b) $t - t_{IP} \geq 1/k_{PBR}$: $3k_{PBR} \exp(-k_{SR}(t - t_{IP})) \left[2\left(\frac{k_{PBR}}{k_{SR}}\right)^2 \left(\exp\left(\frac{k_{SR}}{k_{PBR}}\right) - 1 \right) - \left(1 + 2\frac{k_{PBR}}{k_{SR}}\right) \right]$	(19)

^aVariables: t : reaction time; t_{IP} : time interval of the IP; k_{SR} : rate constant for the SR; and k_{PBR} : rate constant for the PBR.

the reaction interface as long as the degree of the distribution is constant for the kinetic rate data used for the kinetic calculation, and the apparent E_a values do not vary with α .²⁴ Therefore, the constant apparent E_a values observed in this study for a wide α range of $0.1 \leq \alpha \leq 0.9$ for both the α - and β -phases can be interpreted as those for the reaction interface advancements. The good fit of the R_n model to the latter part of the reactions can also be taken as evidence for interface shrinkage in each particle.

3.5. Morphological Changes in the Sample and Physicogeometric Reaction Model. Figure 10 shows the surface textures of partially dehydrated α -phase samples for different α values under isothermal conditions. At the beginning of the reaction, the product phase appears on the surfaces and expands by radiating from the central points with holes (Figure 10a). It is thus apparent that the reaction initiates on the surfaces by nucleation and growth. As the reaction advances, the surfaces become covered with the surface product layers that have small cracks (Figure 10b). The cracks are possible diffusion channels for the removal of water vapor generated at the internal reaction sites. During the SR, the reaction interface likely also advances toward the center of the particles. The shape of each reactant particle is preserved during the course of the reaction (Figure 10c). The surface cracks do not significantly change during the latter half of the reaction.

Figure 11 shows the surface textures of partially dehydrated β -phase samples for different α values under isothermal conditions. Right after the IP, bulges are observed on the surfaces, which show nucleation and growth of the anhydrous product (Figure 11a). During the main reaction stage, large cracks radiating from the edges of quadratic prisms in the direction of the long axes are significant (Figure 11b). The cracks develop further as the reaction advances (Figure 11c). The large cracks expose the internal reaction sites to the atmosphere, possibly making diffusional removal of the water vapor easier.

For both α - and β -phase samples, morphological observations of the partially reacted sample particles indicated a surface-induced process. In addition, for both samples, there was a distribution in the initiation time of the SR among the sample particles. Thus, the distribution of the degree of conversion among the sample particles in the assemblage must be considered for the kinetic interpretation. The difference in surface crack textures between the phases should mean different reaction conditions during advancement of the reaction interface toward the center of the particles. The diffusional removal of water vapor evolved at the reaction interface should be more difficult for the α -phase samples because of the smaller cracks on the surface product layer (Figure 10b). In this situation, there might be a higher water vapor pressure at the reaction interface for the α -phase compared to that at the β -phase. This difference, as well as the differences in crystallographic structure between the α - and β -phase samples, should be considered when interpreting the larger apparent E_a for the α -phase for $0.1 \leq \alpha \leq 0.9$ (Figure 8). Because the outer shape of the sample particles of the α -phase is preserved during the reaction, the diffusion of water vapor through the surface product layer would become even more difficult as the reaction advances. This might explain why, for α -phase samples only, the apparent E_a deviated at the final stage of the reaction.

3.6. Combined Model for the IP–SR–Phase Boundary Reaction (PBR). From the observations of the mass-loss behavior and morphological changes of the samples during thermal dehydration, the reaction is evidently a combined process of IP, SR, and PBR. To interpret the physicogeometrical characteristics of the reaction, a model of the phase-boundary-controlled reaction with a distribution of the degree of conversion due to a distribution of the reaction initiation time for each particle can be applied. As discussed above, a similar reaction model was proposed by Yoshioka et al.³³ for the thermal decomposition of magnesium hydroxide by assuming a random nucleation on the edge surfaces of the hexagonal plate particles and two-dimensional shrinkage of the

reaction interface. Using the model, they derived an integral kinetic equation under isothermal conditions with separate kinetic rate constants for surface nucleation and for PBR. The kinetic equation was adopted by Masuda et al.³⁴ to account for the sigmoidal mass-loss curves for the thermal dehydration of magnesium oxalate dihydrate. The kinetic model has been extended in this study by considering the IP and introducing the dimension of the reaction interface shrinkage n ($n = 1-3$). A series of differential kinetic equations, eqs 14–19 listed in Table 2, were derived for describing the combined IP–SR–PBR reaction processes under isothermal conditions. The detailed derivations of the kinetic equations are described in the Supporting Information. Although the kinetic equations under nonisothermal conditions are necessary for describing all kinetic rate data recorded in this study, the derivation requires some more considerations and is left for a future task.

Using different kinetic equations for different values of n , we found optimum values for t_{IP} ; the rate constant for the nucleation process, k_{SR} ; and the rate constant for the reaction interface advancement, k_{PBR} , by optimization using a nonlinear least-squares analysis minimizing the sum of the squares of the residuals when fitting the experimental curve of $(d\alpha/dt)_{exp}$ versus time with the calculated curve $(d\alpha/dt)_{cal}$ versus time according to the kinetic equations.

$$F = \sum_{j=1}^M \left[\left(\frac{d\alpha}{dt} \right)_{exp,j} - \left(\frac{d\alpha}{dt} \right)_{cal,j} \right]^2 \quad (20)$$

For the kinetic analysis through parameter optimization, the experimentally determined value of t_{IP} was used as the default value. The order of the default k_{PBR} value was deduced from the formal kinetic analysis described above. After setting the default t_{IP} and k_{PBR} values, the order of the default k_{SR} value was determined by graphically comparing the experimental and calculated data.

A typical result for kinetic simulation based on the kinetic equations in Table 2 is shown in Figure 12, as exemplified by those for the thermal dehydration of the β -phase sample at 394 K. Irrespective of the sample phases and reaction temperatures, satisfactory fits of the experimental kinetic curves were obtained when n was set to 2 or 3. Fits using $n = 1$ were inferior. The optimized t_{IP} did not change with the assumed kinetic equation and was practically identical to the experimental t_{IP} shown in Figure 7. The optimized values for k_{SR} and k_{PBR} are summarized in Table S2 (Supporting Information). Thus, the sigmoidal shape of the experimental kinetic curves was adequately described by the IP–SR–PBR model. Using the optimized parameters at different temperatures, the apparent Arrhenius parameters for the respective reaction stages of IP, SR, and PBR were determined. Figure 13 shows the Arrhenius plots assuming $n = 2$. Because of the practically identical values of the optimized and experimentally determined t_{IP} , the apparent values of E_{IP} should be identical to those determined experimentally (Figure 7). Table 3 lists the optimized kinetic parameters for the SR and PBR processes. The values for the apparent Arrhenius parameters, E_{SR} and $\ln A_{SR}$, for the SR processes of the α - and β -phases are nearly identical and do not change with the assumed kinetic equation, except for the β -phase reaction using the kinetic equation with $n = 2$. On the other hand, the Arrhenius parameters for PBR processes are different for the α - and β -phases, although the values do not seem to depend on n . The apparent E_{PBR} value for the PBR process for α -phase is larger than that determined using the

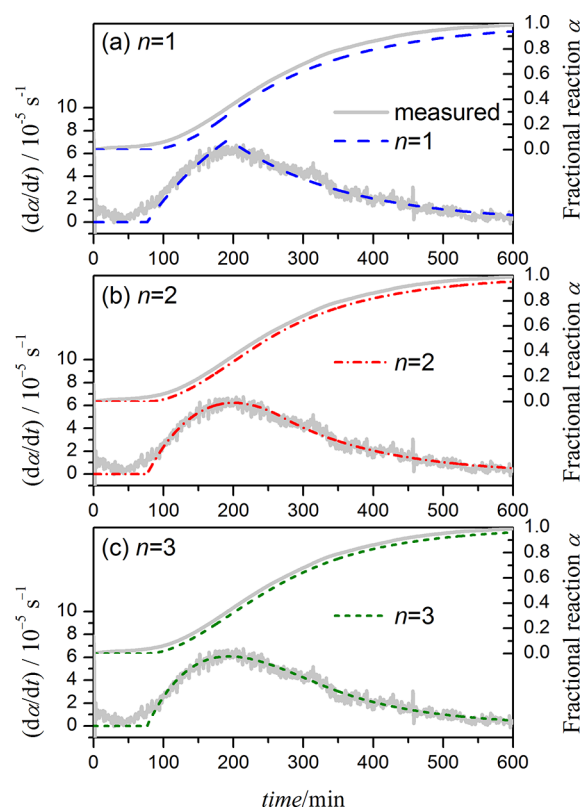


Figure 12. Graphical representation of the fitting results for isothermal dehydration at 394 K of β -FeC₂O₄·2H₂O by the combined IP–SR–PBR model.

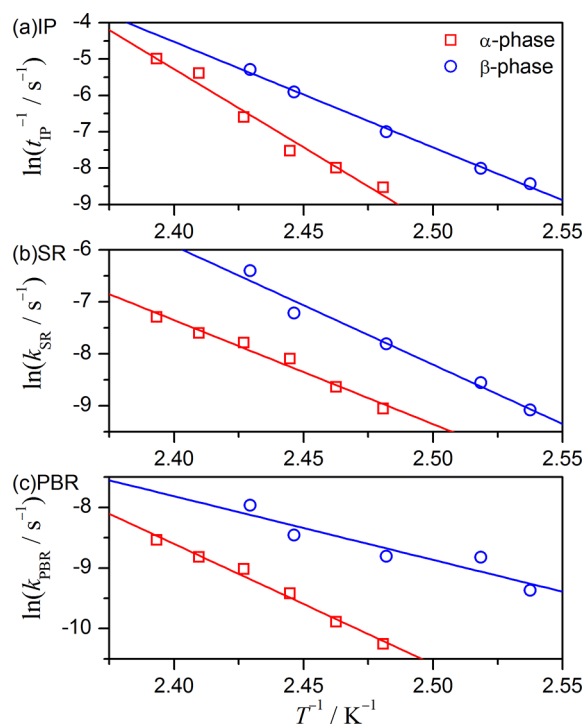


Figure 13. Arrhenius plots for the respective reaction steps for the thermal dehydration of α - and β -FeC₂O₄·2H₂O when assuming $n = 2$ in the combined IP–SR–PBR model.

Friedman method, but the apparent E_{PBR} values for the β -phase are nearly identical to those found using the Friedman method

Table 3. Optimized Kinetic Parameters for SR and PBR Processes for the Isothermal Dehydration of α - and β -FeC₂O₄·2H₂O by Assuming Combined Kinetic Equations with Different Values of n

sample	n	SR			PBR		
		$E_{\text{SR}}/\text{kJ mol}^{-1}$	$\ln(A_{\text{SR}}/\text{s}^{-1})$	γ^a	$E_{\text{PBR}}/\text{kJ mol}^{-1}$	$\ln(A_{\text{PBR}}/\text{s}^{-1})$	γ^a
α -phase	1	176.4 ± 13.5	43.8 ± 3.9	−0.9885	162.1 ± 11.9	38.5 ± 3.5	−0.9893
	2	166.3 ± 12.8	40.7 ± 3.8	−0.9883	165.1 ± 9.6	39.1 ± 2.8	−0.9932
	3	162.1 ± 9.8	39.3 ± 2.9	−0.9927	168.8 ± 11.6	39.9 ± 3.4	−0.9907
β -phase	1	166.5 ± 3.9	41.7 ± 1.2	−0.9992	95.9 ± 28.4	20.5 ± 8.5	−0.8899
	2	190.0 ± 16.3	48.9 ± 4.9	−0.9891	87.3 ± 19.2	17.4 ± 5.7	−0.9343
	3	166.4 ± 15.6	41.8 ± 4.7	−0.9871	104.6 ± 17.9	22.2 ± 5.3	−0.9588

^aCorrelation coefficient of the linear regression analysis of the Arrhenius plot.

within a standard deviation. Regardless of these differences, larger apparent E_{PBR} values for the α -phase in comparison to those for the β -phase are in agreement with the results of the isoconversional kinetic analysis for the overall reactions (Figure 8).

From the kinetic simulation assuming the combined IP–SR–PBR model, the differences in the kinetic behaviors for the thermal dehydration of the α - and β -phases can be explained as follows: (1) The formation of possible nucleation sites on the surfaces of the reactant particles is more difficult for the α -phase, as evidenced by a longer IP when the sample was heated at a constant temperature and by a higher temperature for reaction initiation when the sample was heated at a constant β . A difference in surface morphology could be one reason for the difference, in addition to the crystallographic difference. (2) The SR kinetics assuming random nucleation among different sample particles indicate no significant difference between the α - and β -phases. This implies that once the surfaces are saturated with nucleus-forming sites, the nucleation behavior among the sample particles in the assemblage could be described by similar kinetics. (3) The rate of advancement of the reaction interfaces toward the center of the particles is different for the α - and β -phases. The process for the β -phase is characterized by the smaller values of the Arrhenius parameters in comparison to that for the α -phase. Considering the morphological changes during the reaction, it is deduced that formation of large cracks along the long axis of the columnar particles in the β -phase samples makes the reaction being easier due to the easier diffusional removal of the evolved water vapor. In contrast, because the α -phase has a rigid outer shell of surface product layer during the course of the reaction, diffusional removal of the evolved water vapor is more difficult, and there is a higher water vapor pressure at the internal reaction sites. The difference in the water vapor pressure at the reaction interface of the phase-boundary-controlled reaction can be one of the possible causes of the different E_{PBR} values of the α - and β -phases.

From these physicogeometrical kinetic analyses and morphological studies, we anticipate that the overall kinetics of the thermal dehydration of ferrous oxalate dihydrate polymorphs might be greatly affected by atmospheric water vapor conditions. Also, the kinetic behavior of the IP for both α - and β -phases and the kinetics of the reaction interface advancement for the β -phase might be key processes that control the overall kinetic processes under different water vapor pressure conditions. Detailed experimental studies on the effect of atmospheric water vapor on the overall kinetics of the thermal dehydration of ferrous oxalate dihydrate polymorphs will be reported separately.

4. CONCLUSIONS

The thermal dehydration of ferrous oxalate dihydrate polymorphs indicates significant IPs and the subsequent sigmoidal mass-loss trace under isothermal conditions. Thermal dehydration of square bipyramidal α -phase particles has longer IPs and higher thermal stability in comparison with the quadratic prismatic β -phase particles. The apparent activation energies for both IP and thermal dehydration of the α -phase are larger than those for the β -phase. In both samples, the reaction initiates on the surface with a distribution of the initiation time among the particles in the sample assemblage, which results in a distribution of the degree of conversion among the particles during the reaction. The different formations of cracks in the surface product layer of the α - and β -phases could influence the diffusional removal of water vapor produced at the reaction interfaces and generate local reaction conditions. The combined kinetic model of IP–SR–PBR is applicable to describe the overall process of thermal dehydration. The sigmoidal shape of the mass-loss behavior is simulated satisfactorily, and the kinetic parameters for the three component processes are evaluated separately. The differences in the kinetic parameters for the IP and the PBR processes between the α - and β -phase samples conform to the kinetic behavior expected from the formal kinetic analysis of the reaction and the morphological observations. The kinetic simulation based on the combined model is also expected to be a possible method to evaluate the effects of specific reaction conditions to the respective component processes, for instance, the effect of atmospheric water vapor pressure on the thermal dehydration of ferrous oxalate dihydrate.

■ ASSOCIATED CONTENT

Supporting Information

Derivation of kinetic equations for the combined IP–SR–PBR model. Kinetic results of the optimization of rate constants based on the combined IP–SR–PBR model. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Garner, W. E. *Chemistry of the Solid State*; Butterworths: London, 1955.
- (2) Young, D. A. *Decomposition of Solids*; Pergamon Press: Oxford, U.K., 1966.
- (3) Delmon, B. *Introduction à la Cinétique Hétérogène*; Technip: Paris, 1969.
- (4) Šesták, J.; Šatava, V.; Wendlandt, W. W. The Study of Heterogeneous Processes by Thermal Analysis. *Thermochim. Acta* **1973**, *7*, 333–556.
- (5) Barret, P. *Cinétique Hétérogène*. Gauthier-Villars: Paris, 1973.
- (6) Brown, M. E.; Dollimore, D.; Galwey, A. K. *Reactions in the Solid State*; Elsevier: Amsterdam, The Netherlands, 1980.
- (7) Šesták, J. *Thermophysical Properties of Solids: Their Measurements and Theoretical Thermal Analysis*; Elsevier: Amsterdam, The Netherlands, 1984.
- (8) Galwey, A. K.; Brown, M. E. *Thermal Decomposition of Ionic Solids*; Elsevier: Amsterdam, The Netherlands, 1999.
- (9) Galwey, A. K. Structure and Order in Thermal Dehydrations of Crystalline Solids. *Thermochim. Acta* **2000**, *355*, 181–238.
- (10) Koga, N.; Tanaka, H. A Physico-Geometric Approach to the Kinetics of Solid-State Reactions as Exemplified by the Thermal Dehydration and Decomposition of Inorganic Solids. *Thermochim. Acta* **2002**, *388*, 41–61.
- (11) Galwey, A. K.; Koga, N.; Tanaka, H. A Kinetic and Microscopic Investigation of the Thermal Dehydration of Lithium Sulphate Monohydrate. *J. Chem. Soc., Faraday Trans.* **1990**, *86*, 531–537.
- (12) Tanaka, H.; Koga, N.; Galwey, A. K. Thermal Dehydration of Crystalline Hydrates: Microscopic Studies and Introductory Experiments to the Kinetics of Solid-State Reactions. *J. Chem. Educ.* **1995**, *72*, 251–256.
- (13) Zellelow, A. Z.; Kim, K.-H.; Sours, R. E.; Swift, J. A. Solid-State Dehydration of Uric Acid Dihydrate. *Cryst. Growth Des.* **2010**, *10*, 418–425.
- (14) Koga, N.; Maruta, S.; Kimura, T.; Yamada, S. Phenomenological Kinetics of the Thermal Decomposition of Sodium Hydrogencarbonate. *J. Phys. Chem. A* **2011**, *115*, 14417–14429.
- (15) Koga, N.; Goshi, Y.; Yoshikawa, M.; Tatsuoka, T. Physico-Geometrical Kinetics of Solid-State Reactions in an Undergraduate Thermal Analysis Laboratory. *J. Chem. Educ.* **2014**, *91*, 239–245.
- (16) Šesták, J. Rationale and Fallacy of Thermoanalytical Kinetic Patterns. *J. Therm. Anal. Calorim.* **2011**, *110*, 5–16.
- (17) Koga, N.; Tanaka, H. Kinetics and Mechanisms of the Thermal Dehydration of Dilithium Sulfate Monohydrate. *J. Phys. Chem.* **1989**, *93*, 7793–7798.
- (18) Favregeon, L.; Pijolat, M.; Valdivieso, F.; Helbert, C. Experimental Study and Monte-Carlo Simulation of the Nucleation and Growth Processes During the Dehydration of $\text{Li}_2\text{SO}_4 \cdot \text{H}_2\text{O}$ Single Crystals. *Phys. Chem. Chem. Phys.* **2005**, *7*, 3723–3727.
- (19) Tanaka, H.; Koga, N. Kinetics of the Thermal Dehydration of Potassium Copper(II) Chloride Dihydrate. *J. Phys. Chem.* **1988**, *92*, 7023–7029.
- (20) Koga, N.; Tanaka, H. Kinetic and Morphological Studies of the Thermal Dehydration of α -Nickel(II) Sulfate Hexahydrate. *J. Phys. Chem.* **1994**, *98*, 10521–10528.
- (21) Wada, T.; Koga, N. Kinetics and Mechanism of the Thermal Decomposition of Sodium Percarbonate: Role of the Surface Product Layer. *J. Phys. Chem. A* **2013**, *117*, 1880–1889.
- (22) Koga, N.; Yamada, S.; Kimura, T. Thermal Decomposition of Silver Carbonate: Phenomenology and Physicogeometrical Kinetics. *J. Phys. Chem. C* **2013**, *117*, 326–336.
- (23) Koga, N.; Suzuki, Y.; Tatsuoka, T. Thermal Dehydration of Magnesium Acetate Tetrahydrate: Formation and In Situ Crystallization of Anhydrous Glass. *J. Phys. Chem. B* **2012**, *116*, 14477–14486.
- (24) Koga, N.; Criado, J. M. Kinetic Analyses of Solid-State Reactions with a Particle-Size Distribution. *J. Am. Ceram. Soc.* **1998**, *81*, 2901–2909.
- (25) Tanaka, H.; Koga, N. Self-Cooling Effect on the Kinetics of Nonisothermal Dehydration of Lithium Sulfate Monohydrate. *J. Therm. Anal.* **1990**, *36*, 2601–2610.
- (26) Koga, N.; Criado, J. M. The Influence of Mass Transfer Phenomena on the Kinetic Analysis for the Thermal Decomposition of Calcium Carbonate by Constant Rate Thermal Analysis (CRTA) under Vacuum. *Int. J. Chem. Kinet.* **1998**, *30*, 737–744.
- (27) Koga, N.; Kimizu, T. Thermal Decomposition of Indium(III) Hydroxide Prepared by the Microwave-Assisted Hydrothermal Method. *J. Am. Ceram. Soc.* **2008**, *91*, 4052–4058.
- (28) Favregeon, L.; Pijolat, M.; Helbert, C. A Mechanism of Nucleation During Thermal Decomposition of Solids. *J. Mater. Sci.* **2008**, *43*, 4675–4683.
- (29) Kimura, T.; Koga, N. Thermal Dehydration of Monohydrocalcite: Overall Kinetics and Physico-Geometrical Mechanisms. *J. Phys. Chem. A* **2011**, *115*, 10491–10501.
- (30) Mampel, K. L. Time Conversion Formulas for Heterogeneous Reactions at the Phase Boundaries of Solid Bodies, I: The Development of the Mathematical Method and the Derivation of Area Conversion Formulas. *Z. Phys. Chem., Abt. A* **1940**, *187*, 43–57.
- (31) Viricelle, J.-P.; Pijolat, M.; Soustelle, M.; Zing, C. Transformation of Cerium(III) Hydroxycarbonate into Ceria. Part 1. Nucleation and Growth Rates of Ceria. *J. Chem. Soc., Faraday Trans.* **1995**, *91*, 4431–4435.
- (32) Bouineau, V.; Pijolat, M.; Soustelle, M. Characterisation of the Chemical Reactivity of a CaCO_3 Powder for Its Decomposition. *J. Eur. Ceram. Soc.* **1998**, *18*, 1319–1324.
- (33) Yoshioka, H.; Amita, K.; Hashizume, G. The Nucleation-Two Dimensional Interface Growth Equation for the Thermal Decomposition of $\text{Mg}(\text{OH})_2$. *Netsu Sokutei* **1984**, *11*, 115–118.
- (34) Masuda, Y.; Iwata, K.; Ito, R.; Ito, Y. Kinetics of the Thermal Dehydration of Magnesium Oxalate Dihydrate in a Flowing Atmosphere of Dry Nitrogen. *J. Phys. Chem.* **1987**, *91*, 6543–6547.
- (35) Pijolat, M.; Brun, C.; Valdivieso, F.; Soustelle, M. Reduction of Uranium Oxide U_3O_8 to UO_2 by Hydrogen. *Solid State Ionics* **1997**, *101–103*, 931–935.
- (36) Valdivieso, F.; Bouineau, V.; Pijolat, M.; Soustelle, M. Kinetic Study of the Dehydration of Lithium Sulphate Monohydrate. *Solid State Ionics* **1997**, *101–103*, 1299–1303.
- (37) Pijolat, M.; Favregeon, L.; Soustelle, M. From the Drawbacks of the Arrhenius- $f(\alpha)$ Rate Equation Towards a More General Formalism and New Models for the Kinetic Analysis of Solid–Gas Reactions. *Thermochim. Acta* **2011**, *525*, 93–102.
- (38) Koga, N.; Sato, Y. Formation and Transformation Kinetics of Amorphous Iron(III) Oxide during the Thermally Induced Transformation of Ferrous Oxalate Dihydrate in Air. *J. Phys. Chem. A* **2011**, *115*, 141–151.
- (39) Brown, R. A.; Bevan, S. C. The Thermal Decomposition of Ferrous Oxalate Dihydrate. *J. Inorg. Nucl. Chem.* **1966**, *28*, 387–391.
- (40) Macklen, E. D. Influence of Atmosphere on the Thermal Decomposition of Ferrous Oxalate Dihydrate. *J. Inorg. Nucl. Chem.* **1967**, *29*, 1229–1234.
- (41) Rao, V.; Shashimohan, A. L.; Biswas, A. B. Studies on the Formation of $\gamma\text{-Fe}_2\text{O}_3$ (Maghemite) by Thermal Decomposition of Ferrous Oxalate Dihydrate. *J. Mater. Sci.* **1974**, *9*, 430–433.
- (42) Rane, K. S.; Nikumbh, A. K.; Mukhedkar, A. J. Thermal Decomposition of Ferrous Oxalate Dihydrate Studied by Direct Current Electrical Conductivity Measurements. *J. Mater. Sci.* **1981**, *16*, 2387–2397.
- (43) Boyanov, B.; Khadzhiev, D.; Vasilev, V. Study of Thermal Decomposition of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$. *Thermochim. Acta* **1985**, *93*, 89–92.
- (44) Mohamed, M. A.; Galwey, A. K. A Kinetic and Mechanistic Study of the Isothermal Decomposition of Ferrous Oxalate Dihydrate. *Thermochim. Acta* **1993**, *213*, 269–278.
- (45) Chhabra, V.; Lal, M.; Maitra, A. N.; Ayyub, P. Preparation of Ultrafine High Density Gamma Ferric Oxide Using Aerosol OT Microemulsions and its Characterization. *Colloid Polym. Sci.* **1995**, *273*, 939–946.

- (46) Rupard, R. G.; Gallagher, P. K. The Thermal Decomposition of Coprecipitates and Physical Mixtures of Magnesium–Iron Oxalates. *Thermochim. Acta* **1996**, 272, 11–26.
- (47) Carles, V.; Alphonse, P.; Tailhades, P.; Rousset, A. Study of Thermal Decomposition of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ under Hydrogen. *Thermochim. Acta* **1999**, 334, 107–113.
- (48) Popa, M.; Calderon-Moreno, J. M.; Crisan, D.; Zaharescu, M. Study of Fe Addition on the Thermal Decomposition of Coprecipitated Oxalates for the Bi-Based Superconductor Synthesis. *J. Therm. Anal. Calorim.* **2000**, 62, 633–645.
- (49) Frost, R. L.; Weier, M. L. Thermal Decomposition of Humboldtine: A High Resolution Thermogravimetric and Hot Stage Raman Spectroscopic Study. *J. Therm. Anal. Calorim.* **2004**, 75, 277–291.
- (50) Hermanek, M.; Zboril, R.; Mashlan, M.; Machala, L.; Schneeweiss, O. Thermal Behaviour of Iron(II) Oxalate Dihydrate in the Atmosphere of Its Conversion Gases. *J. Mater. Chem.* **2006**, 16, 1273–1280.
- (51) Hermanek, M.; Zboril, R.; Medrik, I.; Pechousek, J.; Gregor, C. Catalytic Efficiency of Iron(III) Oxides in Decomposition of Hydrogen Peroxide: Competition between the Surface Area and Crystallinity of Nanoparticles. *J. Am. Chem. Soc.* **2007**, 129, 10929–10936.
- (52) Angermann, A.; Töpfer, J. Synthesis of Magnetite Nanoparticles by Thermal Decomposition of Ferrous Oxalate Dihydrate. *J. Mater. Sci.* **2008**, 43, 5123–5130.
- (53) Hermankova, P.; Hermanek, M.; Zboril, R. Thermal Decomposition of Ferric Oxalate Tetrahydrate in Oxidative and Inert Atmospheres: The Role of Ferrous Oxalate as an Intermediate. *Eur. J. Inorg. Chem.* **2010**, 2010, 1110–1118.
- (54) Sorensen, O. T.; Rouquerol, J. *Sample Controlled Thermal Analysis*; Kluwer: Dordrecht, The Netherlands, 2003.
- (55) Perez-Maqueda, L. A.; Criado, J. M.; Gotor, F. J. Controlled Rate Thermal Analysis Commanded by Mass Spectrometry for Studying the Kinetics of Thermal Decomposition of Very Stable Solids. *Int. J. Chem. Kinet.* **2002**, 34, 184–192.
- (56) Deyrieux, R.; Peneloux, A. Contribution à l'étude des Oxalates de Certains Métaux Bivalents: Structure Cristalline des Deux Formes Allotropiques de l'oxalate Ferreux Dihydraté. *Bull. Soc. Chim. France* **1969**, 8, 2675–2681.
- (57) Bickley, R. I.; Edwards, H. G. M.; Rose, S. J. A Raman Spectroscopic Study of Nickel(II) Oxalate Dihydrate, $\text{NiC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ and Dipotassium Bisoxalatonicel(II) Hexahydrate, $\text{K}_2\text{Ni}(\text{C}_2\text{O}_4)_2 \cdot 6\text{H}_2\text{O}$. *J. Mol. Struct.* **1991**, 243, 341–350.
- (58) Frost, R. L.; Weier, M. L. Raman Spectroscopy of Natural Oxalates at 298 and 77 K. *J. Raman Spectrosc.* **2003**, 34, 776–785.
- (59) Frost, R. L.; Locke, A.; Martens, W. N. Synthesis and Raman Spectroscopic Characterisation of the Oxalate Mineral Wheatleyite $\text{Na}_2\text{Cu}^{2+}(\text{C}_2\text{O}_4)_2 \cdot 2\text{H}_2\text{O}$. *J. Raman Spectrosc.* **2008**, 39, 901–908.
- (60) Criado, J. M.; Ortega, A.; Gotor, F. Correlation between the Shape of Controlled-Rate Thermal Analysis Curves and the Kinetics of Solid-State Reactions. *Thermochim. Acta* **1990**, 157, 171–179.
- (61) Pérez-Maqueda, L. A.; Ortega, A.; Criado, J. M. The Use of Master Plots for Discriminating the Kinetic Model of Solid State Reactions from a Single Constant-Rate Thermal Analysis (CRTA) Experiment. *Thermochim. Acta* **1996**, 277, 165–173.
- (62) Koga, N. Ozawa's Kinetic Method for Analyzing Thermoanalytical Curves: History and Theoretical Fundamentals. *J. Therm. Anal. Calorim.* **2013**, 113, 1527–1541.
- (63) Friedman, H. L. Kinetics of Thermal Degradation of Char-Forming Plastics from Thermogravimetry, Application to a Phenolic Plastic. *J. Polym. Sci., Part C: Polym. Chem.* **1964**, 6, 183–195.
- (64) Ozawa, T. Applicability of Friedman Plot. *J. Therm. Anal.* **1986**, 31, 547–551.
- (65) Koga, N. Kinetic Analysis of Thermoanalytical Data by Extrapolating to Infinite Temperature. *Thermochim. Acta* **1995**, 258, 145–159.
- (66) Gotor, F. J.; Criado, J. M.; Málek, J.; Koga, N. Kinetic Analysis of Solid-State Reactions: The Universality of Master Plots for Analyzing Isothermal and Nonisothermal Experiments. *J. Phys. Chem. A* **2000**, 104, 10777–10782.
- (67) Ozawa, T. A New Method of Analyzing Thermogravimetric Data. *Bull. Chem. Soc. Jpn.* **1965**, 38, 1881–1886.
- (68) Ozawa, T. Non-Isothermal Kinetics and Generalized Time. *Thermochim. Acta* **1986**, 100, 109–118.
- (69) Šesták, J.; Berggren, G. Study of the Kinetics of the Mechanism of Solid-State Reactions at Increasing Temperatures. *Thermochim. Acta* **1971**, 3, 1–12.
- (70) Šesták, J. Diagnostic Limits of Phenomenological Kinetic Models Introducing the Accommodation Function. *J. Therm. Anal.* **1990**, 36, 1997–2007.
- (71) Avrami, M. Kinetics of Phase Change. I. General Theory. *J. Chem. Phys.* **1939**, 7, 1103–1112.
- (72) Avrami, M. Kinetics of Phase Change. II. Transformation–Time Relations for Random Distribution of Nuclei. *J. Chem. Phys.* **1940**, 8, 212–223.
- (73) Avrami, M. Kinetics of Phase Change. III. Granulation, Phase Change, and Microstructure. *J. Chem. Phys.* **1941**, 9, 177–184.
- (74) Barmak, K. A Commentary On: “Reaction Kinetics in Processes of Nucleation and Growth”. *Metall. Mater. Trans. A* **2010**, 41, 2711–2775.
- (75) Málek, J.; Criado, J. M. Empirical Kinetic Models in Thermal Analysis. *Thermochim. Acta* **1992**, 203, 25–30.
- (76) Perez-Maqueda, L. A.; Criado, J. M.; Sanchez-Jimenez, P. E. Combined Kinetic Analysis of Solid-State Reactions: A Powerful Tool for the Simultaneous Determination of Kinetic Parameters and the Kinetic Model without Previous Assumptions on the Reaction Mechanism. *J. Phys. Chem. A* **2006**, 110, 12456–12462.
- (77) Koga, N.; Tanaka, H. Accommodation of the Actual Solid-State Process in the Kinetic Model Function. 1. Significance of the Nonintegral Kinetic Exponents. *J. Therm. Anal.* **1994**, 41, 455–469.
- (78) Finney, E. E.; Finke, R. G. Is There a Minimal Chemical Mechanism Underlying Classical Avrami–Erofe'ev Treatments of Phase-Transformation Kinetic Data? *Chem. Mater.* **2009**, 21, 4692–4705.